

Grammatical Moods in Maths Writing

A The grammatical mood (*спосіб дієслова*) is a feature of verbs, used for indicating the intentions of the speaker. Most of the mathematics is written in one of the following moods: *the indicative mood*, *the imperative mood*, and *the interrogative mood*. These are all finite forms of the verb. Infinitives, gerunds, and participles, which are non-finite forms of the verb, are not considered to be examples of moods.

The **indicative mood** (*дійсний спосіб*) expresses factual, non-hypothetical information:

- The group G **is** abelian.

The **imperative mood** (*наказовий спосіб*) issues instructions:

- **Let** G be an abelian group.

The **interrogative mood** (*питальний спосіб*) asks questions:

- **Is** the group G abelian?

B The other moods and types of moods occur rarely in formal maths writing. They convey a degree of uncertainty and therefore are uncommon in a discipline that centres on proving universal truths.

The **conditional mood** (*умовний спосіб*) is sometimes used in maths writing to express counterfactual possibility.

- Since both $R \times$ and BR^+ are abelian and the product of abelian groups is abelian, then the group G **would be** abelian. But this is not true and gives a contradiction.

The **subjunctive mood** (*підрядний спосіб*) is used to discuss imaginary or hypothetical situations. This mood would only appear in a discussion that explored alternative proofs or options.

- If the group G **were** abelian, then the result would have been true.
(*But given that G is not abelian, the result may not be true.*)

C The subjunctive mood in the example above should be distinguished from the typical *if ..., then ...* pattern that uses the present tense indicative to state a fact.

- If the group G **is** abelian, then the result **is** true.

Alternative cue words to represent the *if ..., then ...* statement in mathematics are as follows:

If p , then q

p when q
 p implies q
 p only if q
 q , if p
 q , if and only if p
 q , assuming that p
 q , given that p
provided that p , q

If not p , then q

q unless p
unless p , q
until p , q
without p , q
 q , else p

D If more than one mood is used in the same sentence, a shift in the mood occurs:

- **Prove** (*Imperative*) that G **is** abelian. (*Indicative*)